

Welcome

AI for Global Health

Introduction

LiGHT

Laboratory for intelligent
Global Health & Humanitarian
Response Technologies.

EPFL

ARIADNE LABS


ASHOKA
UNIVERSITY

**WORLD
ECONOMIC
FORUM**

Rwanda
Centre for the
Fourth Industrial
Revolution

Welcome

AI for Global Health

Introduction

LiGHT

Laboratory for intelligent
Global Health & Humanitarian
Response Technologies.

EPFL

ARIADNE LABS


ASHOKA
UNIVERSITY

**WORLD
ECONOMIC
FORUM**

Rwanda
Centre for the
Fourth Industrial
Revolution

Welcome

BiGHT - Building Intelligent Global Health and Humanitarian response technologies

Introduction

LiGHT

Laboratory for intelligent
Global Health & Humanitarian
Response Technologies.

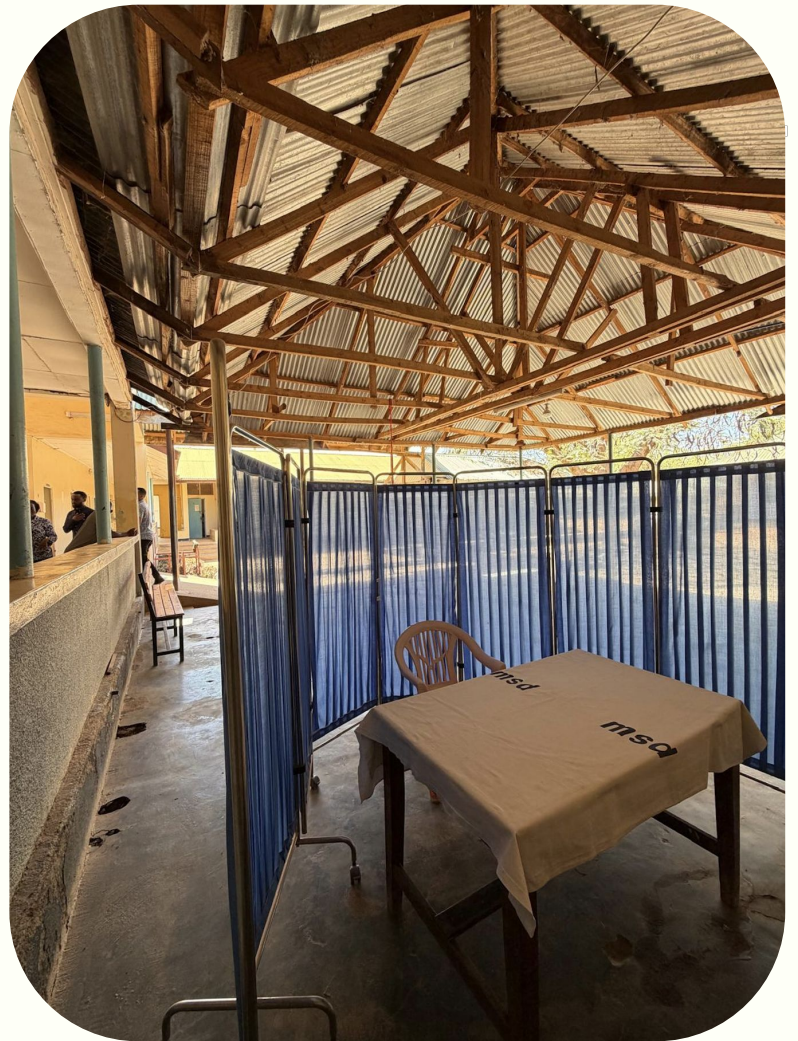
EPFL

ARIADNE LABS

ASHOKA
UNIVERSITY

**WORLD
ECONOMIC
FORUM**

Rwanda
Centre for the
Fourth Industrial
Revolution

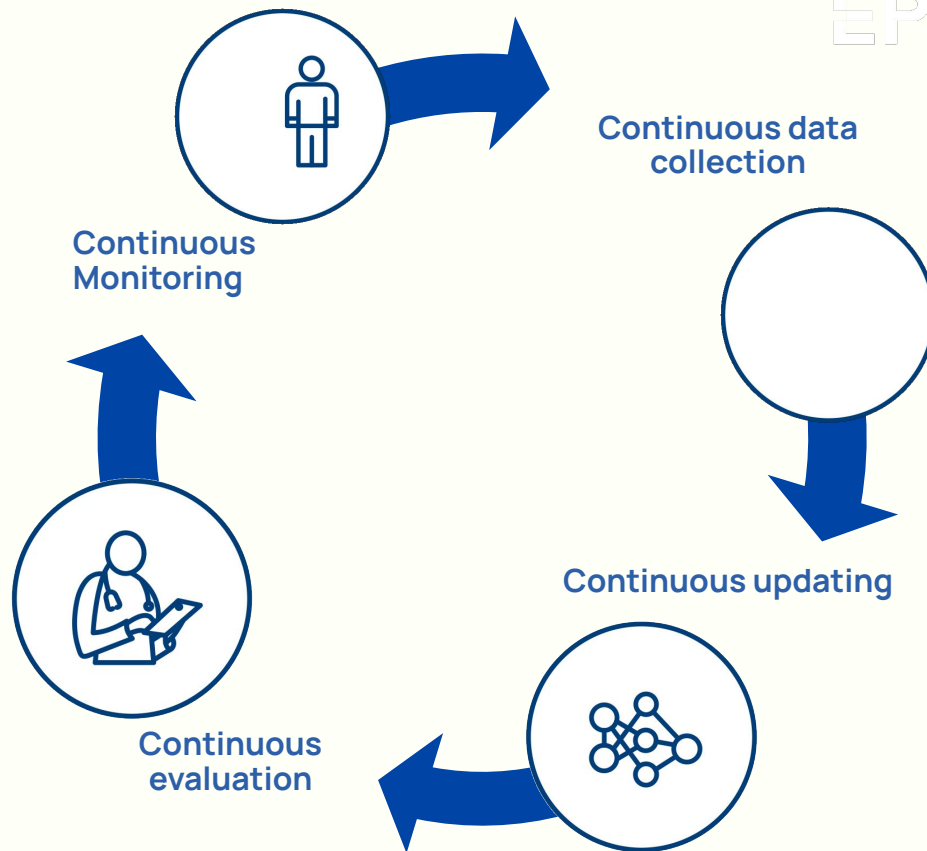




Welcome

EPFL

How we work



Welcome

EPFL

Where we focus



Welcome

What we make

Our work combines strong engineering, evaluation, and implementation science to build AI that works in the real world.

(meditronFO, MOOVE, MoBayes, DISCO, AntibioGo)

LiGHT

Laboratory for intelligent
Global Health & Humanitarian
Response Technologies.



Welcome

Who we are



Annie (Mary-Anne) Hartley

Director



Kristina Keitel

Clinical Director



Lars Klein

Academic Director



David Sasu

Technology Director

Contact information

Fabrice: fabrice.nemo@epfl.ch
Annie: mary-anne.hartley@epfl.ch



Fabrice Nemo

Laboratory for intelligent
Global Health & Humanitarian Academic Coordinator
Response Technologies.



Yusuf Kesmen

EPFL AI Center PhD Fellow



Jeremy Baffou

External TA



**Xavier
Theimer-Lienhard**

PhD Candidate

LiGHT

Paired lectures

Block	Weeks	Goal
Volatile Contexts	1-4	Understand the environments in which AI must operate, and why technology often fails in humanitarian and clinical settings.
High-stakes Decisions	5-7	Understand how clinicians make decisions under uncertainty and how AI can safely support human expertise.
Trustworthy Evidence	9-13	Design, evaluate, and deploy AI systems that are safe, effective, and ready for real-world use.

Paired lectures

Week	Date	Theme	Context lecture	Engineering lecture
1	Sep 9	Inequitable Inaccuracy	Tutti Fratelli: The Principles That Created a Movement	Building for Broken Environments
2	Sep 16	When the Hospital Is the Target	When the Hospital Is the Target	Systems That Work When Nothing Else Does
3	Sep 23	The Last Mile	Health for All, Care for Few	The Last Mile: Getting Intelligence to the Edge
4	Sep 30	Listening Before the Outbreak	The Map That Stopped an Epidemic	Listening to the World: Epidemic Intelligence at Scale

Paired lectures

Week	Date	Theme	Context lecture	Engineering lecture
5	Oct 7	Models Without Patients	The Machine That Was Right and Was Never Used	From Rules to Reasoning
6	Oct 14	Reasoning Against Ourselves	The Confident Fool: How Expertise Breeds Error	Debiasing the Machine
7	Oct 28	Advice at the Wrong Time	Why Doctors Ignore Good Advice	Integration Without Interruption

Paired lectures

Week	Date	Theme	Context lecture	Engineering lecture
9	Nov 11	The Fragility of Accuracy	How We Learned to Count What We Were Losing	When 99% Accuracy Is Dangerous
10	Nov 18	Causal Evidence in Practice	The Man Who Asked Medicine to Prove Itself	Causal Inference: Beyond Correlation
11	Nov 25	Designing for Failure	The Holes in the Cheese	Uncertainty as Architecture
12	Dec 2	Responsibility in Clinical AI	First, Do No Harm - Then, Prove It	Governing What You Cannot Fully Understand
13	Dec 9	Systems That Reach People	Why Good Ideas Die at the Door	Shipping to Scale: The MLOps of Global Health AI

Course Logistics



Laboratory for intelligent
Global Health & Humanitarian
Response Technologies.

Communication during the semester

The screenshot shows the Moodle interface for the course 'AI for global health / BiGHT'. The top navigation bar includes the EPFL logo and links for 'Accueil', 'Tableau de bord', 'Mes cours', and 'My Media'. The course title is displayed in red, followed by a breadcrumb trail 'Informatique (IN) / IN - Master'. Below the title, a horizontal menu contains 'Cours', 'Paramètres', 'Participants', 'Notes', 'Rapports', and 'Plus'. The main content area is divided into two sections: 'Généralités' and 'Discussion'. The 'Généralités' section includes links for 'Course website', 'Anonymous Feedback Form', and 'Project application' (with a deadline of September 16, 2026), along with a note about adding a GitHub handle. The 'Discussion' section features a link for 'Ed Discussion'.

EPFL Accueil Tableau de bord Mes cours My Media

Informatique (IN) / IN - Master

AI for global health / BiGHT

Cours Paramètres Participants Notes Rapports Plus

Généralités Tout replier

- [Course website](#)
- [Anonymous Feedback Form](#)
- [Project application](#)
S'ouvre : mercredi 9 septembre 2026, 15:00 Se termine : mercredi 16 septembre 2026, 22:00
- Please tell us your GitHub handle to add you to the right repository

Discussion

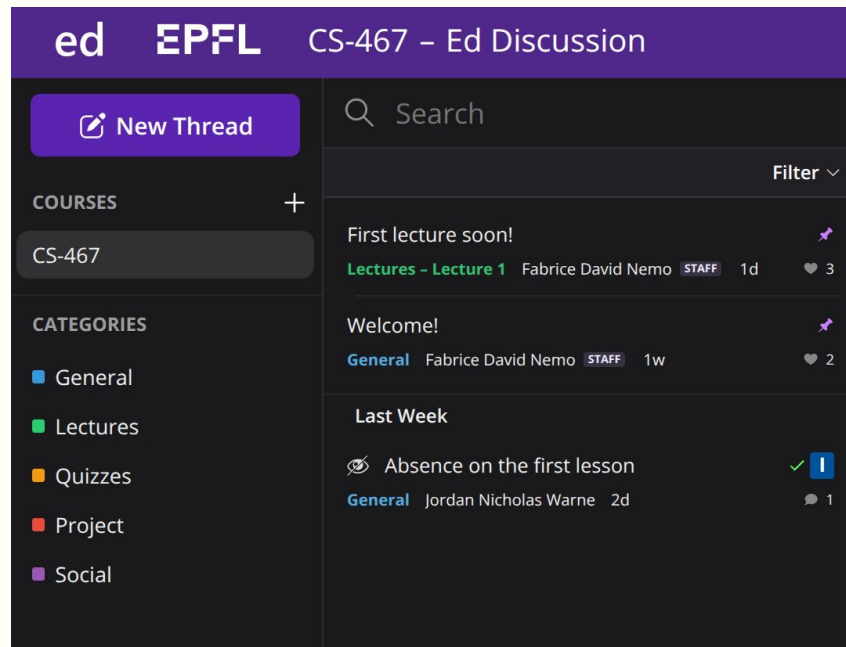
- [Ed Discussion](#)

Communication during the semester

Let's use the ed discussion forum for communication

Use Categories to sort your questions

Check whether your question has already been posted



Website

Course overview

PROJECT
Project Documentation
Project Briefs
Available after Sep 9, 15:00

Anonymous Feedback

BLOCK I - VOLATILE CONTEXTS
Week 1 - Inequitable Inaccuracy
Available after Sep 9, 15:00
Week 2 - When the Hospital Is the Target
Available after Sep 16, 15:00
Week 3 - The Last Mile
Available after Sep 23, 15:00
Week 4 - Listening Before the Outbreak
Available after Sep 30, 15:00

BLOCK II - HIGH-STAKES DECISIONS
Week 5 - Models Without Patients
Available after Oct 7, 15:00
Week 6 - Reasoning Against Ourselves
Available after Oct 14, 15:00
Week 7 - Advice at the Wrong Time
Available after Oct 28, 15:00
Week 8 - Midterm
Available after Nov 4, 15:00

BLOCK III - TRUSTWORTHY EVIDENCE
Week 9 - Fragility of Accuracy
Available after Nov 11, 15:00

LiGHT Building intelligent Global Health and
Humanitarian Response Technologies (BiGHT)



Step 1 of 5

Previous

1. Summary

Next (Course Overview)

Summary

CS-467 Course Curriculum - Fall 2026 - EPFL IC

Summary

This course teaches the principles and practice of designing, building, and rigorously evaluating trustworthy, representative, and contextually appropriate AI systems for high-stakes decision making in global health and humanitarian response.

Lecturers: Prof. Mary-Anne Hartley, Prof. Kristina Keitel, Dr. Lars Klein, Dr. David Sasu, Tim Arni, Yusuf Kesmen, Fabrice Nemo

Coordinator: Fabrice Nemo

TAs: Yusuf Kesmen, Jérémy Baffou

Website

Week 1: Inequitable Inaccuracy

1. Overview ▾
1. Overview
2. Learning Objectives
3. Core Ideas and Case Anchor
4. Why Inequity Produces Inaccuracy
5. Humanitarian Principles as Design Constraints

Step 1 of 10

Previous

1. Overview ▾

Next (Learning Objectives)

Overview

Block I: Volatile Contexts (Weeks 1-4)

Central question: Why do systems fail the people who need them most?

Big idea: Inequity is not background noise. It decides who appears in our data, who reaches care, and who is harmed when we optimize for the people who are already visible.

Block focus: Understand the environments in which AI must operate, and why technology often fails in humanitarian and clinical settings.

Giving Feedback



Anonymous Feedback 

BiGHT Feedback Form

This form lets you send the course team anonymous feedback, so that the course can be improved even before the official EPFL indicative feedback form.

fabrice.nemo@epfl.ch [Changer de compte](#)



 Non partagé

Overall, how satisfied are you with the course so far?

I hate it 1 2 3 4 5 I love it

☐ ☐ ☐ ☐ ☐

How useful have the lectures been for helping you understand the stakes in Global Health / Humanitarian Response and the engineering means for tackling them?

Not useful, I barely learnt anything and had to search for other sources 1 2 3 4 5 The lectures were quite informative and I could effectively learn from them

☐ ☐ ☐ ☐ ☐

What should we keep, change, add, or remove from the lectures?

Real world project

Please stay for the exercise session for a presentation of each project, you will apply and make groups by the end of next week

Project number	Project title	Team size	Short description
Project 1	Offline Speech-To-Speech System For Humanitarian Triage	 5	Offline speech-to-speech system that can help humanitarian workers in Yemen with triage.
Project 2	Interactive Public Health Messenger	5	Build an interactive public health messenger that helps people understand crisis instructions, checks whether guidance was understood, and aggregates misunderstanding patterns for responders.
Project 3	Geospatial Intelligence For Public Health Decisions	5	The main aim of the project is to provide reliable Geospatial Intelligence for Public Health decisions.
Project 4	AI-Powered Disease Risk Prediction System For Zoonotic Disease Outbreaks	 5	The aim of this project is to build an AI-powered disease risk prediction system that integrates human, animal, and environmental data to identify and visualize areas at high risk of zoonotic disease outbreaks.

Grading

Component	Weight
Milestone 1: Technical Design and Reproducible Repository	10%
Milestone 2: Proof Of Concept	20%
Midterm oral presentation	10%
Final presentation with a live demo	10%
Project report and code	30%
In-class tests and quizzes (top 5 of the 6 quizzes)	20%

Questions, so far?

Quick poll

Rather have deadlines on Fridays or Sundays?

Our treat

**Our treat: a field trip to the ICRC
Museum in Geneva, *covered by*
*the lab***

Date TBD soon

LiGHT

Laboratory for intelligent
Global Health & Humanitarian
Response Technologies.



Building intelligent Global Health and Humanitarian
Response Technologies

Week 1: *Inequitable Inaccuracy*

Context *Tutti Fratelli*: The Principles That Created a Movement

LiGHT

Laboratory for intelligent
Global Health & Humanitarian
Response Technologies.

EPFL

ARIADNE LABS

ASHOKA
UNIVERSITY

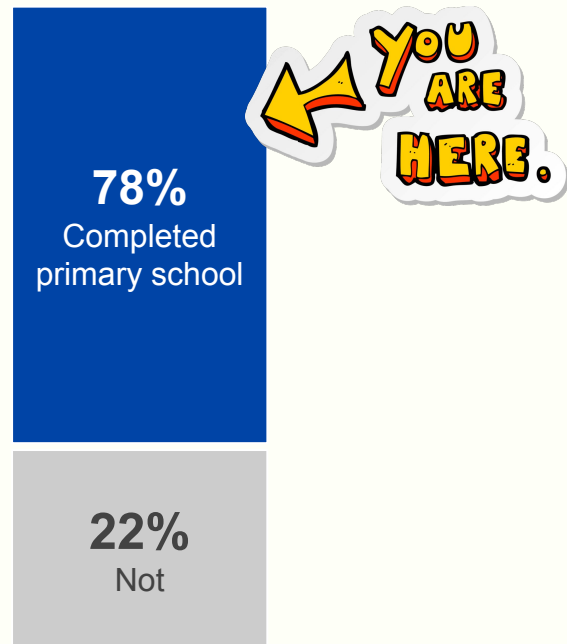
**WORLD
ECONOMIC
FORUM**

Rwanda
Centre for the
Fourth Industrial
Revolution

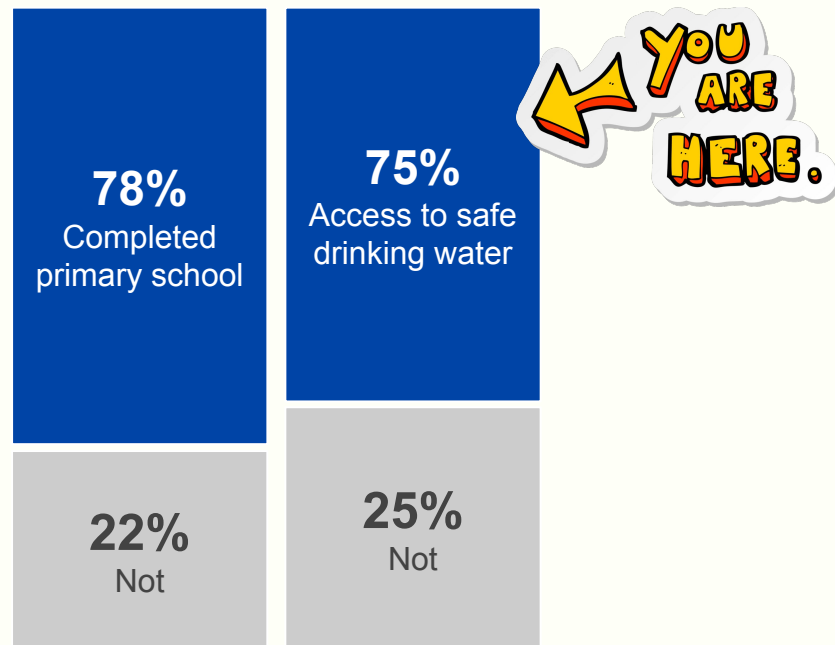
Why do systems *fail* the people who need them most?

- A model that assumes connectivity has already chosen its users.
- A dataset that requires a clinic visit has already excluded the sickest.
- An objective function that maximizes throughput has already decided whose time matters.

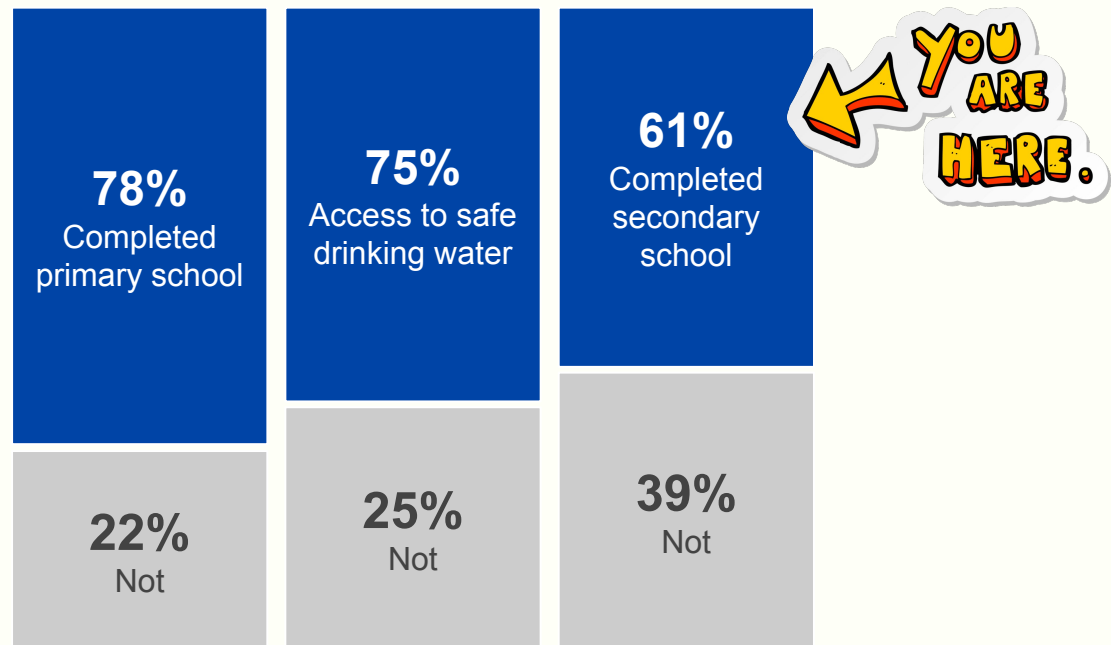
Welcome to the ELITE!



Welcome to the ELITE!



Welcome to the ELITE!



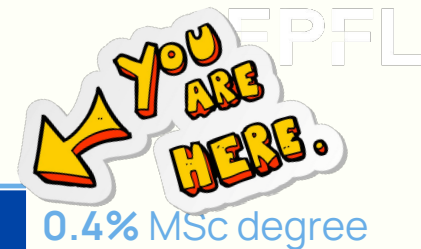
Welcome to the ELITE!



Welcome to the ELITE!

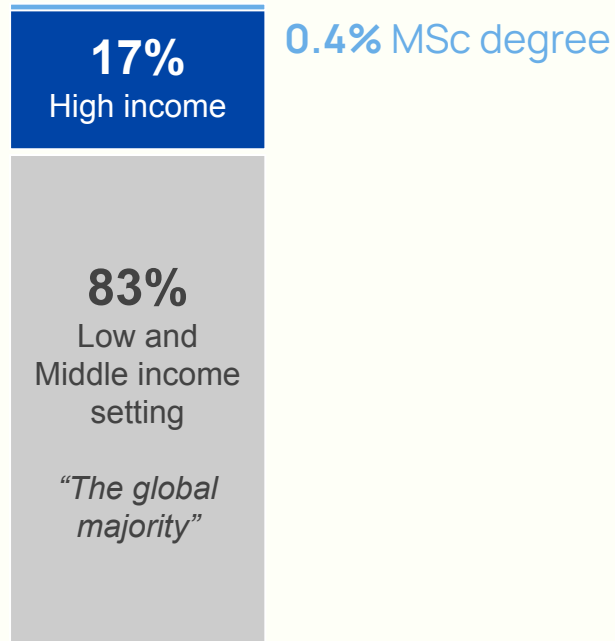


Welcome to the ELITE!

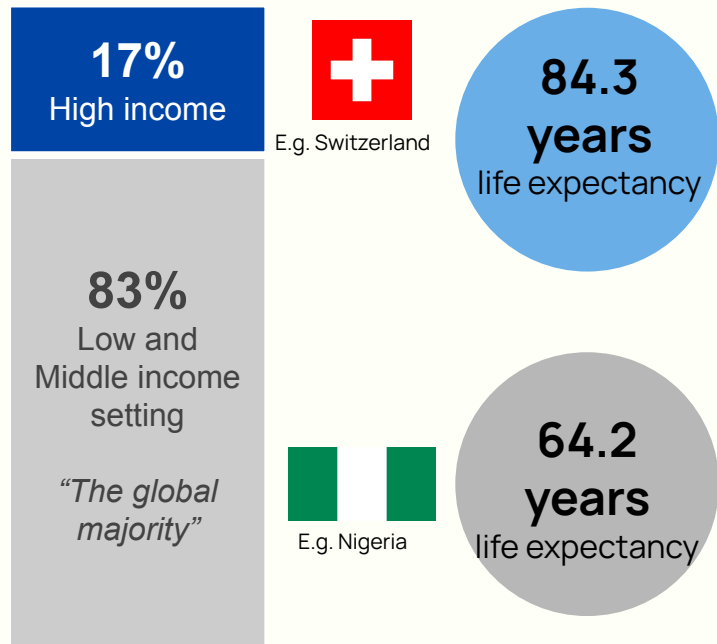


0.4% MSc degree

Welcome to the ELITE! What will you do with this privilege?

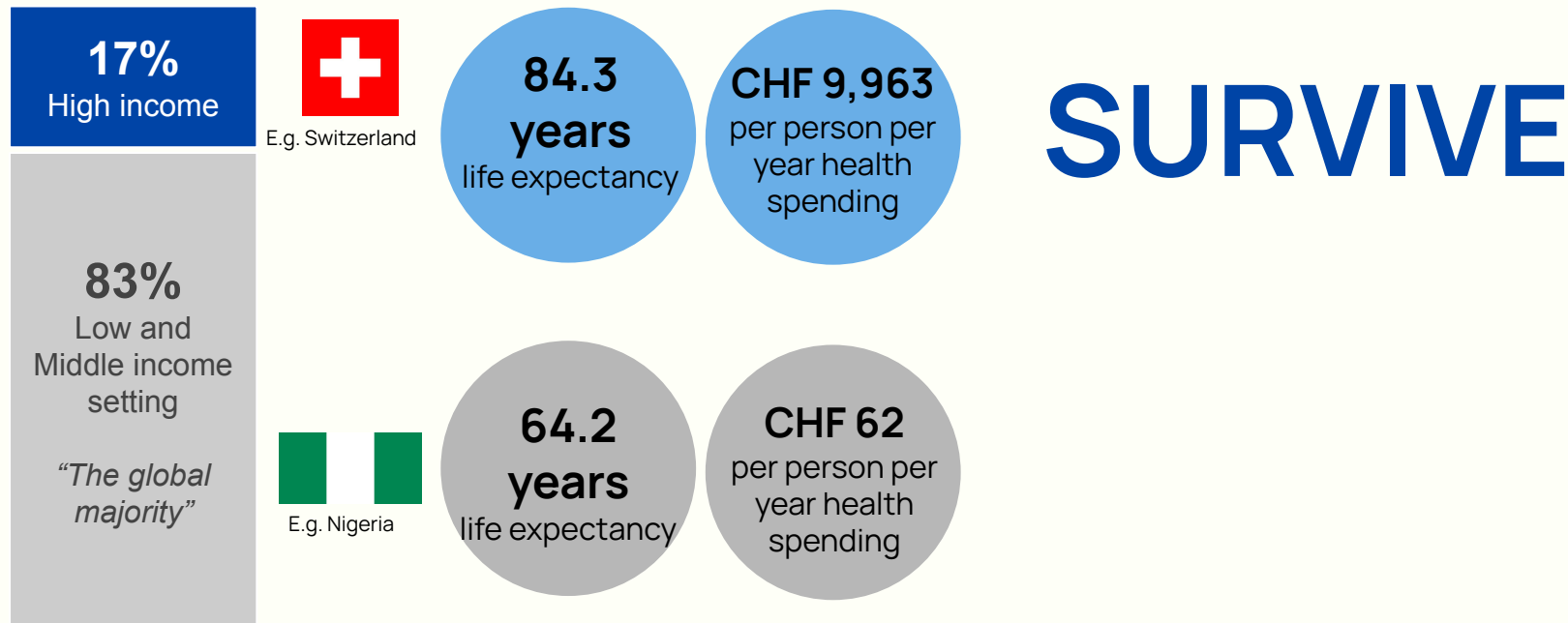


Welcome to the ELITE! What will you do with this privilege?

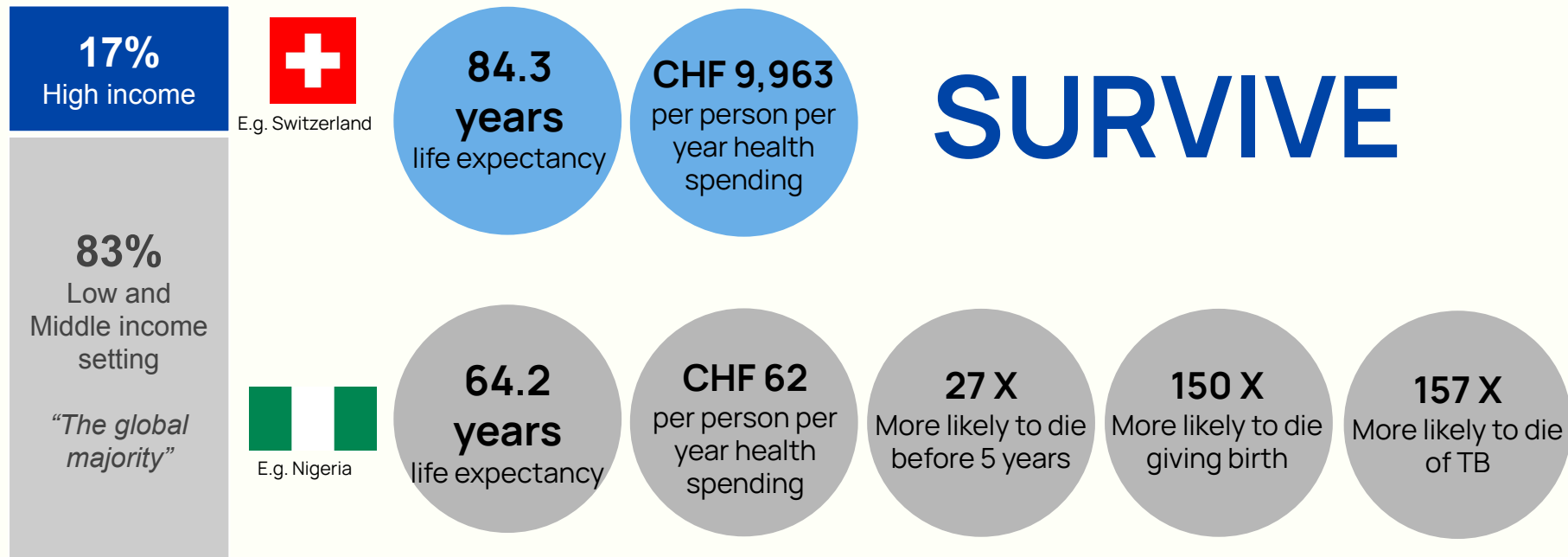


SURVIVE

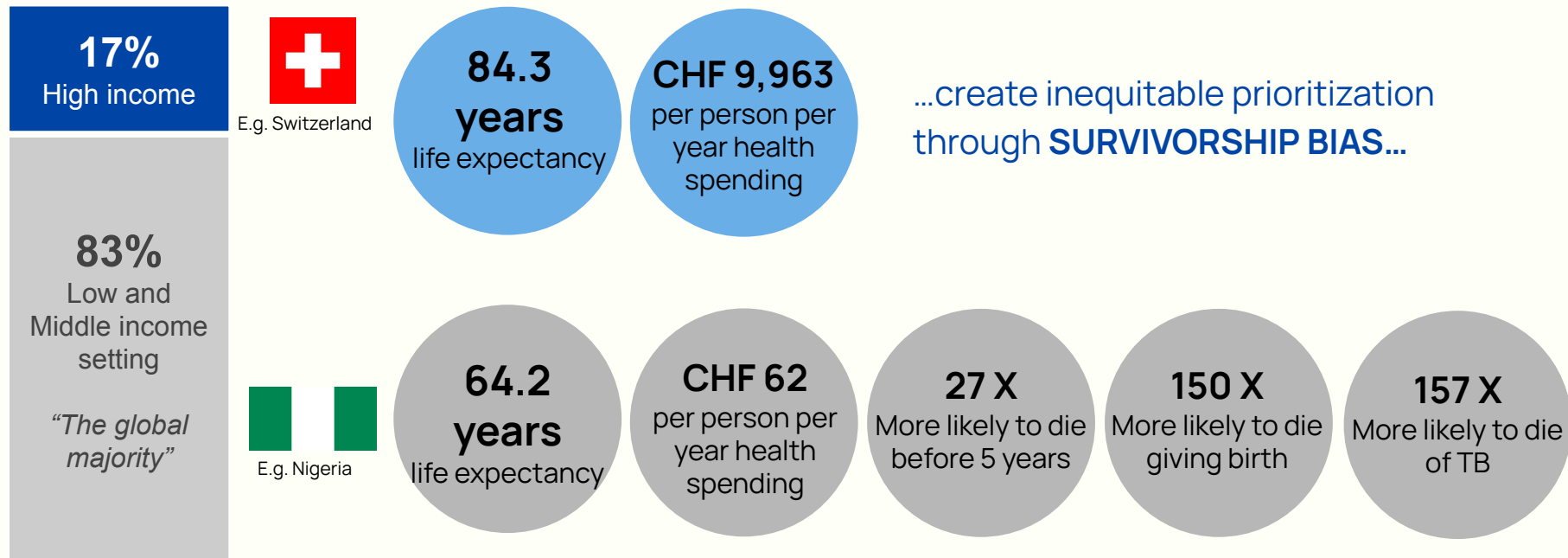
Welcome to the ELITE! What will you do with this privilege?



Welcome to the ELITE! What will you do with this privilege?



Welcome to the ELITE! What will you do with this privilege?



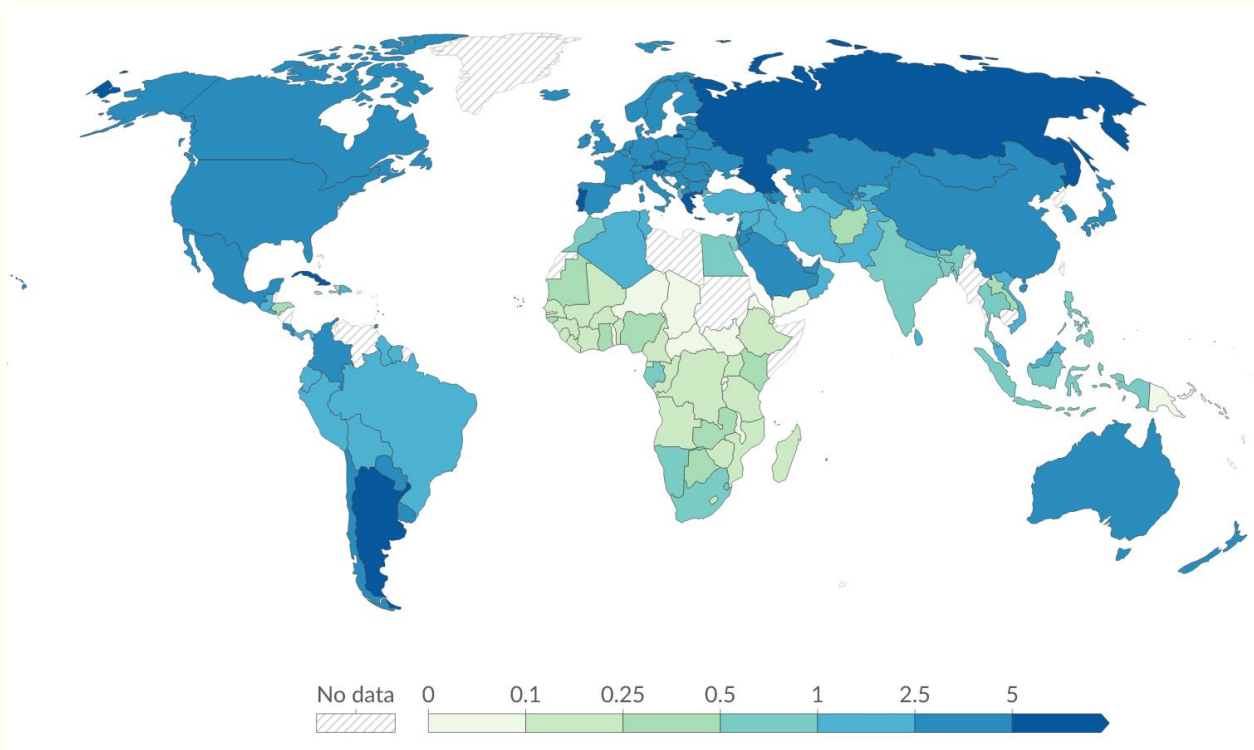
The lottery



1 doctor / 220



1 doctor / 4000



Medical doctors per 1,000 people

The lottery

Gini Index:

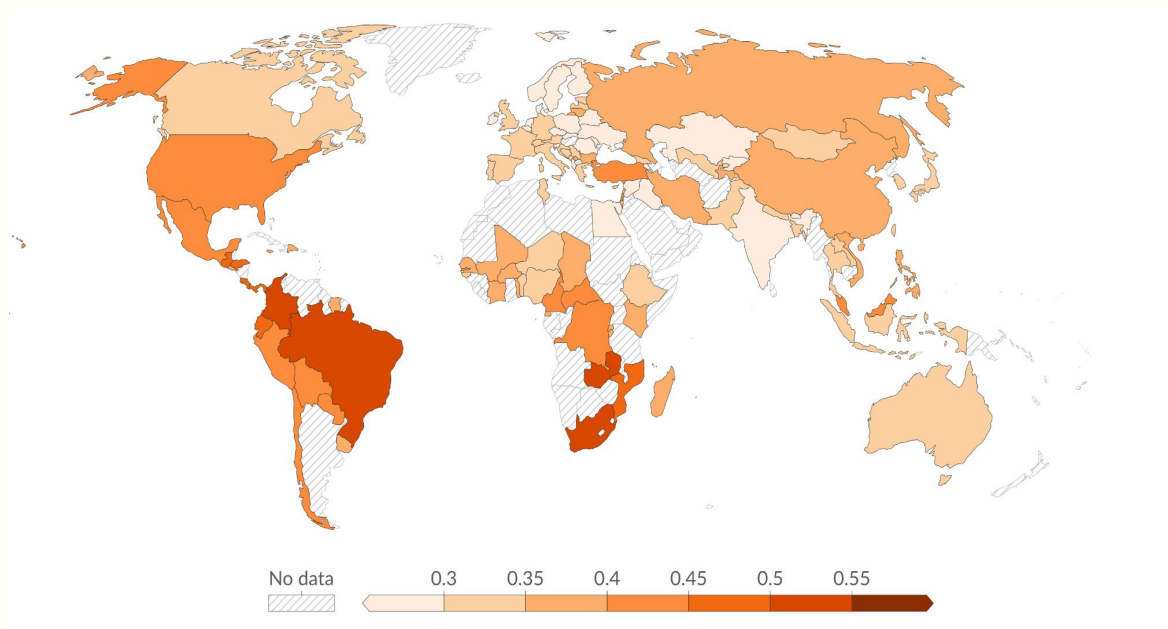
0 (perfect equality)

1 (maximum inequality)



0.6 = Most inequitable country

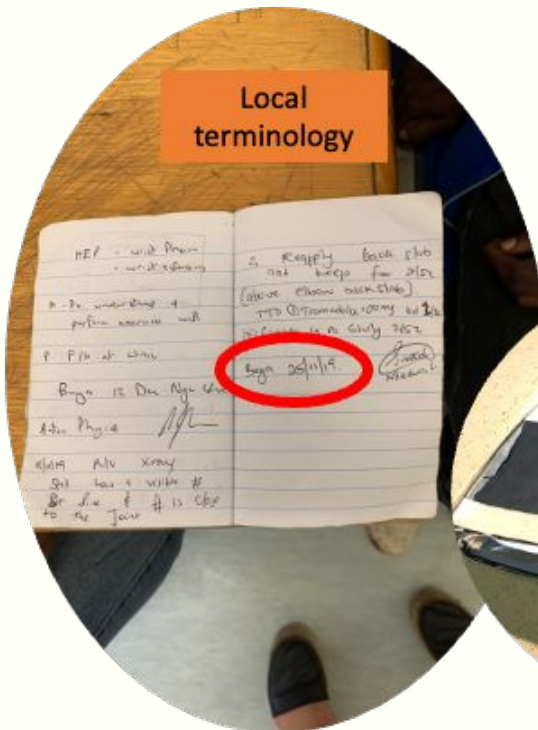
- Richest 10% get 60%
- Poorest 50% get 5%



Data



Patient-owned
health data



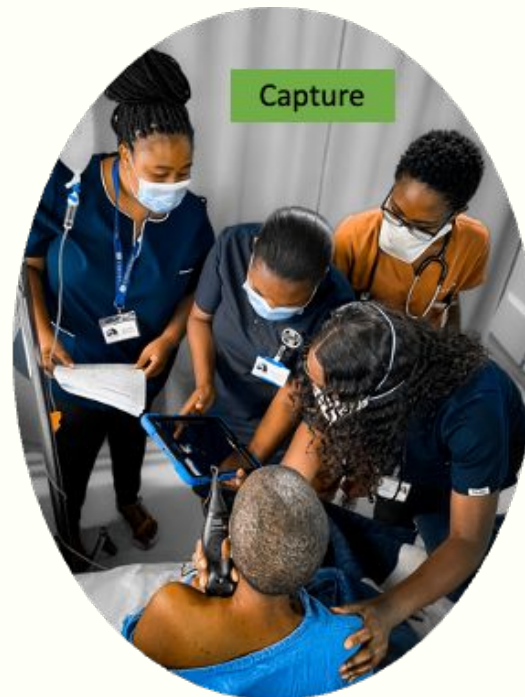
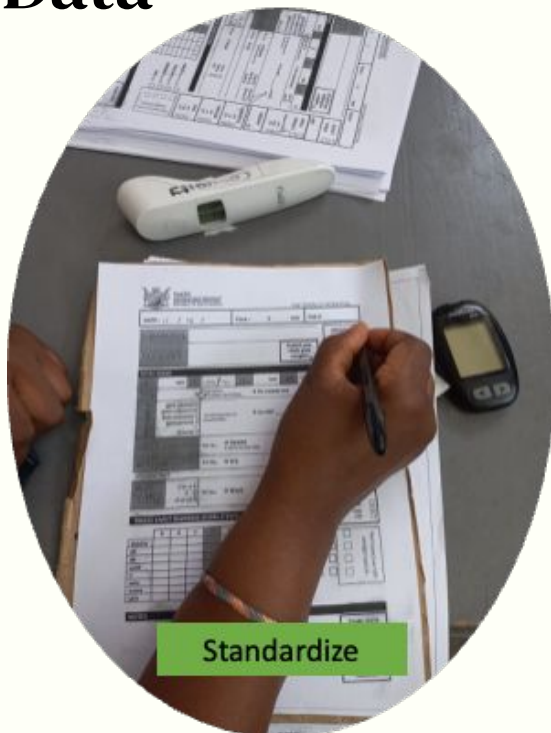
Local
terminology



Reporting to the
government registry

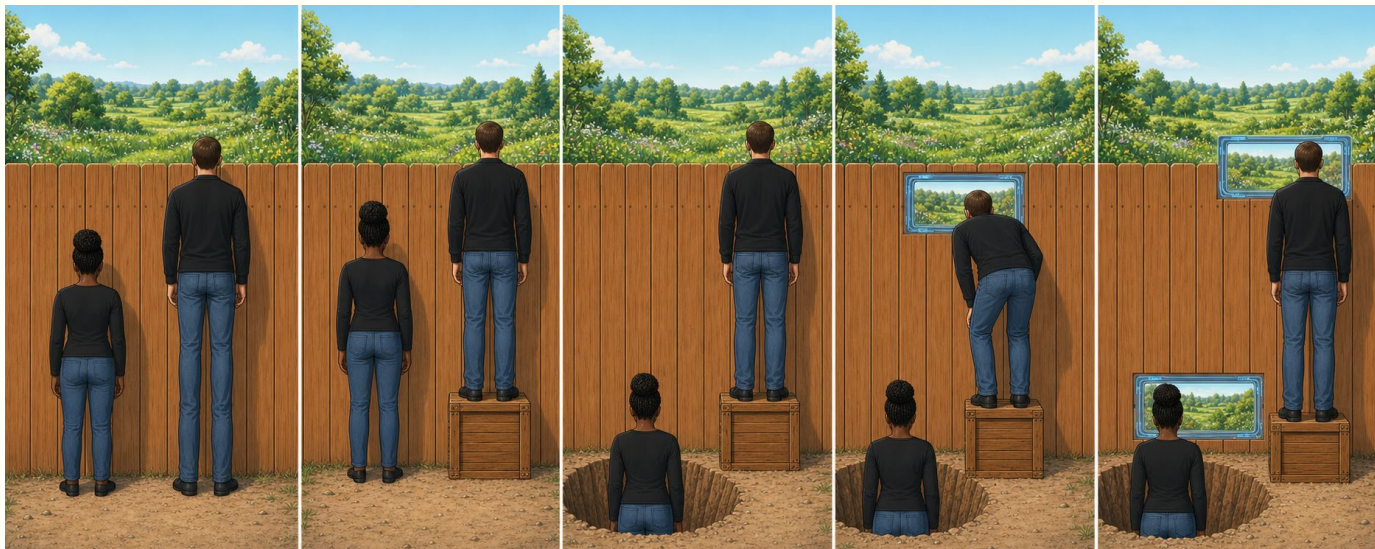
Week 1: An introduction to inequity

Data



Week 1: An introduction to inequity

Equity vs Equality



Innate deficiency /
superiority?

Resource
advantage?

Resource
disadvantage?

Poorly adapted AI

Adapted AI

LiGHT

Laboratory for intelligent
Global Health & Humanitarian
Response Technologies.

Tutti Fratelli: The Principles That Created a Movement

Lecturer: *Prof. Annie Hartley*

Light

Laboratory for intelligent
Global Health & Humanitarian
Response Technologies.

Outline

- 01 The history of humanitarian response
- 02 The humanitarian principles
- 03 The humanitarian system
- 04 Disasters
- 05 Humanitarian standards

Solferino June 24, 1859

CONTEXT



40'000
wounded
soldiers

1 medic /
1000 soldiers

1
veterinarian
/ 100
horses

Henry Dunant

Before the war: Business man from Geneva in the water-mill industry

1859: Travels via Solferino on business...discovers the aftermath

Next days: Organizes relief in the church of Castiglione

Not a medic. Achieves relief through coordination and advocacy

Next years: Founds ICRC

1867: His business goes bankrupt

1901: First Nobel Peace Prize



Founding of the ICRC

1859 (war)

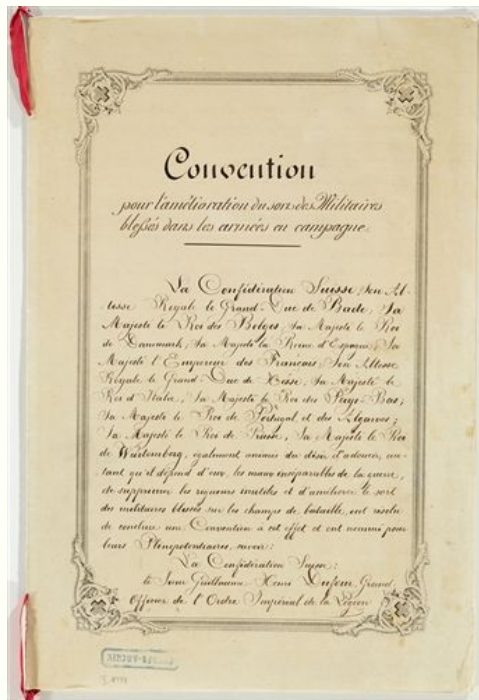
1862 (book)

1863 (ICRC)

1864 (1st Geneva convention)

10 articles (highlights)

- Wounded and sick soldiers must be collected and treated, **regardless of which side they fought for**.
- Medical personnel, ambulances, civilians helping, and military hospitals were declared **neutral** and **protected**.
- The **red cross on a white background** = protective symbol for medical services.



Terminology

Temoignage

“bearing witness”

The humanitarian practice of **speaking out** (publicly or privately) about suffering, violence, neglect, or abuse witnessed during aid work.

Four key terms

Humanity

The claim that the **duty to relieve suffering wherever it is found** overrides other considerations.

Impartiality

No discrimination as to nationality, race, religious belief, class, or political opinion.
Relief is guided solely by need, with priority to the most urgent cases of distress

Neutrality

The Movement **may not take sides** in hostilities or engage at any time in controversies of a political, racial, religious or ideological nature.

Independence

Humanitarian action must remain **autonomous** from political, economic, or military objectives. Able to act on the principles even against the interests of donors and host states.

Applied to AI?

Humanity

Purpose-driven

Use AI only when it protects life, relieves suffering, and preserves human dignity. Efficiency must never outweigh foreseeable harm to people.

Impartiality

Fairness

AI-supported decisions must be based on need alone. Not identity, status, data availability or politics. Test data and outputs for bias, exclusion, and unequal errors.

Neutrality

Interpretability

Humanitarian AI must not take sides or support military targeting, partisan surveillance, propaganda, or political objectives that undermine trust.

Independence

Control

Governments, donors, militaries, and technology companies must not determine whom AI serves. Humanitarian actors must retain meaningful control and the power to question, override, refuse, or stop its use.

Dilemmas

- **Neutrality** may require restraint to preserve trust and access
- **Humanity** may require speaking out against severe suffering
- **Impartiality** may require giving more assistance to those in greatest need even when this looks like taking sides.

Terminology

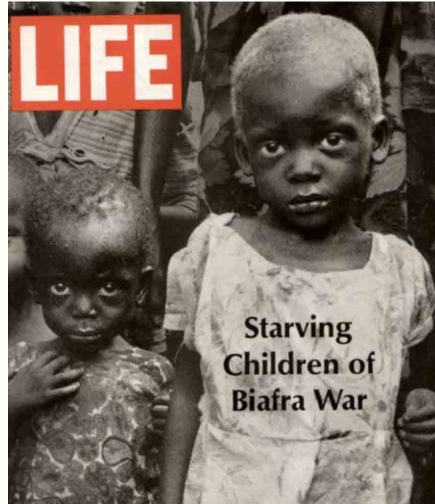
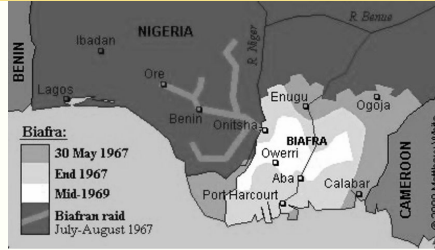
Dilemma

A situation in which two legitimate duties conflict, so that protecting one requires compromising another. No available choice fully satisfies every principle.

Dilemmas that shaped the humanitarian system

1969 (Biafra war)

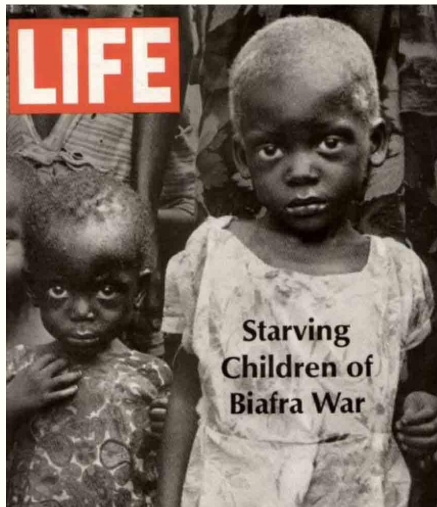
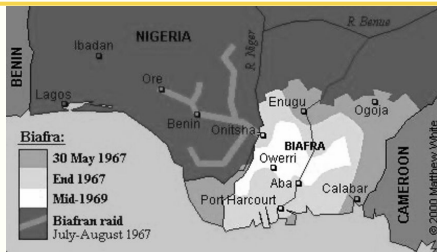
- Biafra succeeds from Nigeria
- Encircled by Nigerian forces
- **Massive famine (2M died)**



Dilemmas that shaped the humanitarian system

1969 (Biafra war)

Biafra succeeds from Nigeria
→ Encircled by Nigerian forces
→ **Massive famine (2M died)**



Dilemma: Access vs Protest

- 1) **Access as protection**
Remain present, negotiate confidentially and preserve access to victims.
- 2) **Protest as protection**
Treat victims, bear public witness and mobilise pressure against intolerable suffering.



ICRC



Dilemmas that shaped the humanitarian system

1969 (Biafra war)



1980 (Vietnam boat people)



Communist victory in Vietnam

→ 100k+ fled to sea ... but no country let them in as refugees

→ A ship with 2K people onboard were starving

Dilemmas that shaped the humanitarian system

1969 (Biafra war)



1980 (Vietnam boat people)



Communist victory in Vietnam

→ 100k+ fled to sea ... but no country let them in as refugees

→ A ship with 2K people onboard were starving

Dilemma: Dialogue vs Action

- 1) **Diplomacy as protection**
Negotiate with states to secure safe landing (but leave leave at risk).
- 2) **Direct action as protection**
Go to those stranded at sea, bear witness and pressure governments to act.



Dilemmas that shaped the humanitarian system

1969 (Biafra war)



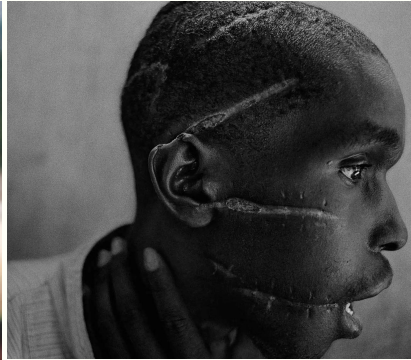
1980 (Vietnam boat people)



1994 (Rwandan Genocide)



People killed
6/minute
330/hour



1M killed in 100 days, 2M fled into neighbouring countries

→ Overcrowded camps lacked sufficient water, sanitation and healthcare

→ Armed groups diverted humanitarian assistance

LiGHT

Laboratory for intelligent
Global Health & Humanitarian
Response Technologies.

Dilemmas that shaped the humanitarian system

1969 (Biafra war)



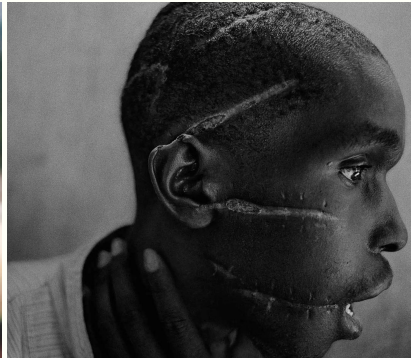
1980 (Vietnam boat people)



1994 (Rwandan Genocide)



People killed
6/minute
330/hour



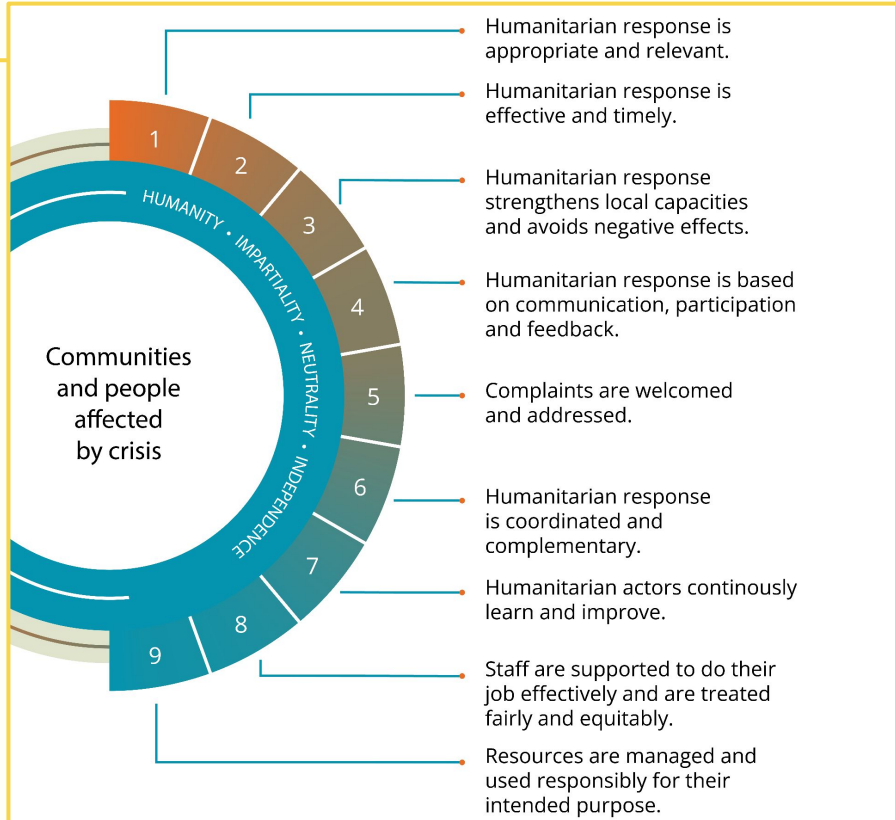
1M killed in 100 days, 2M fled into neighbouring countries
→ Overcrowded camps lacked sufficient water, sanitation and healthcare
→ Armed groups diverted humanitarian assistance

Dilemma: Is any aid better than no aid?

- 1. Urgency as protection**
Deliver whatever assistance is available now –regardless of quality
- 2. Standards as protection**
People have a right to competent and coordinated assistance. Agencies must meet minimum standards and remain accountable for the effects of their aid.



Sphere standards



1994 (Rwanda Genocide)



Humanitarian Charter

+

Measurable Minimum Standards

To ensure sustainable survival and dignity

Sphere standards...what is “minimum”?

WATER



SHORT-TERM EXCEPTION

7.5 L

per person / day
Acute drought • short time only

MAXIMUM DISTANCE

≤ 500 m

From every household
to a water point



SANITATION



RAPID-ONSET

1 : 50

Communal • improve
as soon as possible

MAXIMUM DISTANCE

≤ 50 m

Dwelling to shared toilet

MEDIUM-TERM RATIO

3 : 1

Female : male toilets

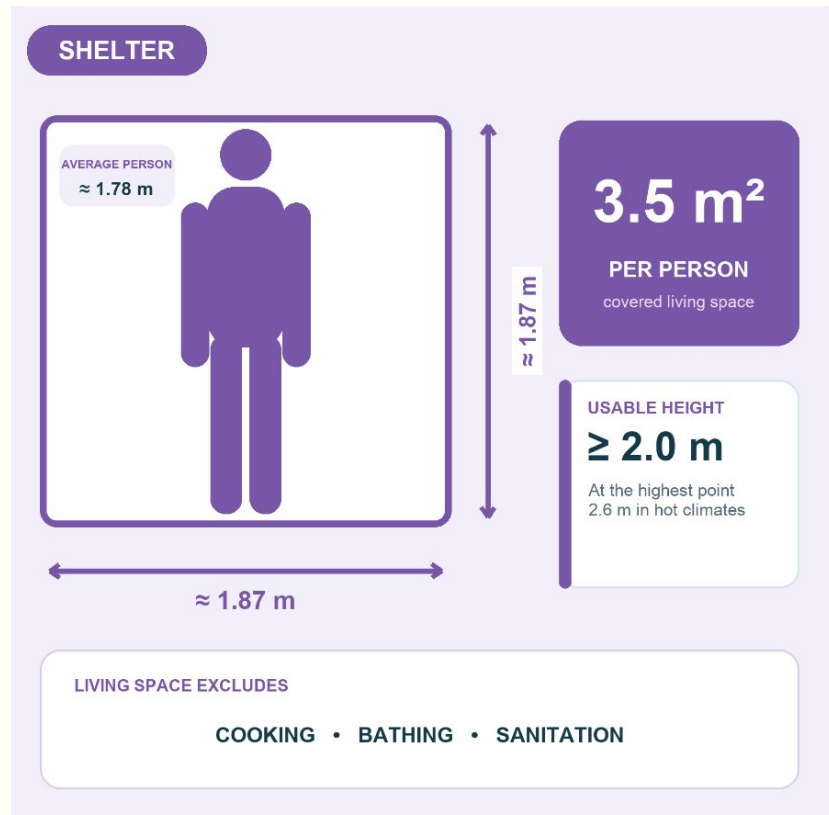
1 TOILET : 20 PEOPLE



SAFETY

Inside locks + adequate lighting
for every shared toilet

Sphere standards...what is “minimum”?



Disaster

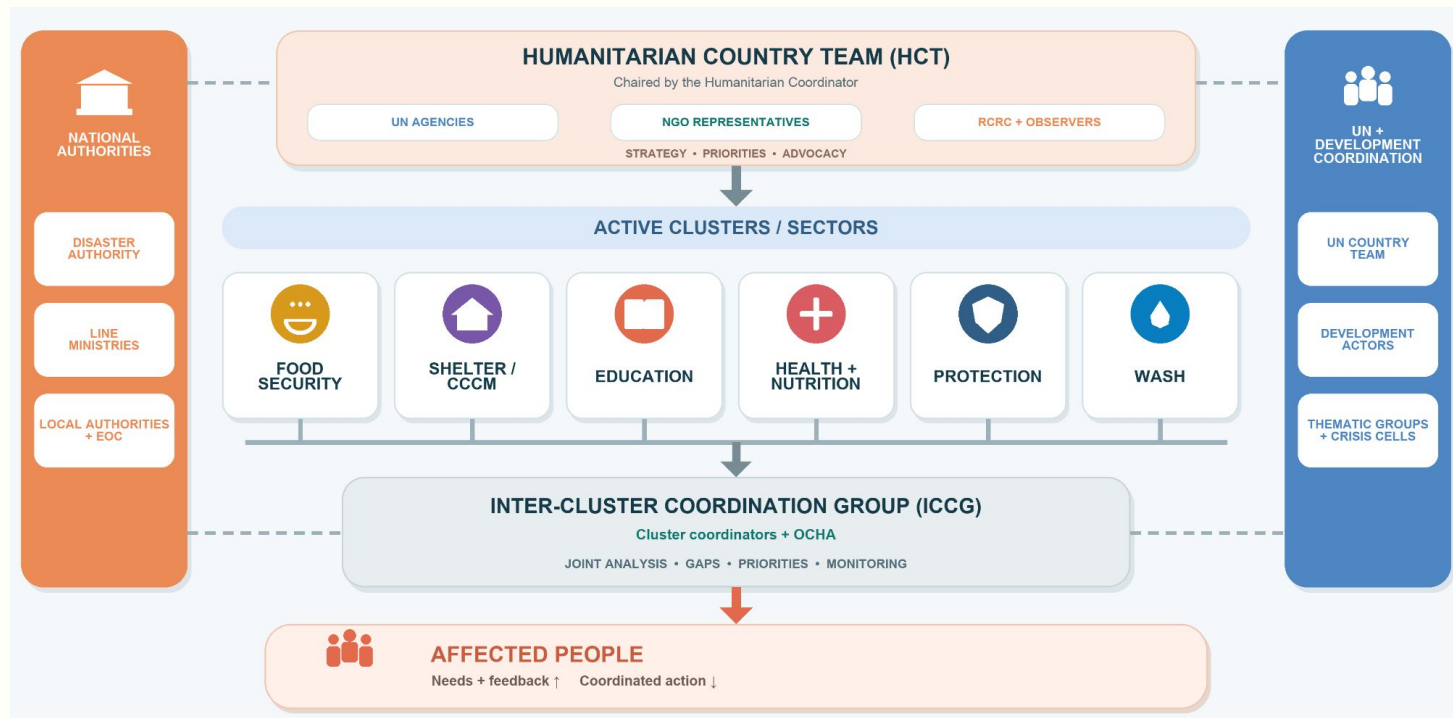
$$\text{Impact} = \frac{\text{Hazard} \times \text{Exposure} \times \text{Vulnerability}}{\text{Capacity}}$$

Terminology

Disaster

A serious disruption of a community's functioning that exceeds its capacity to cope using its own resources.

Coordination



Terminology

Clusters

The UN-led coordination model that divides response into **sectoral clusters**

- Health (WHO)
- Shelter
- Nutrition
- WASH
- Protection
- ...

Data protection is protection

150M records since 1914

Persistence = value

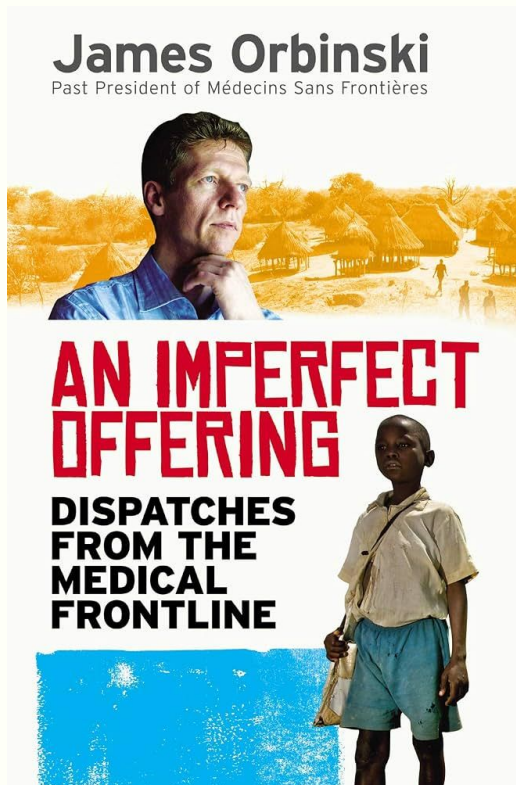
...but also lengthens lifetime of exposure risk



Connecting to AI

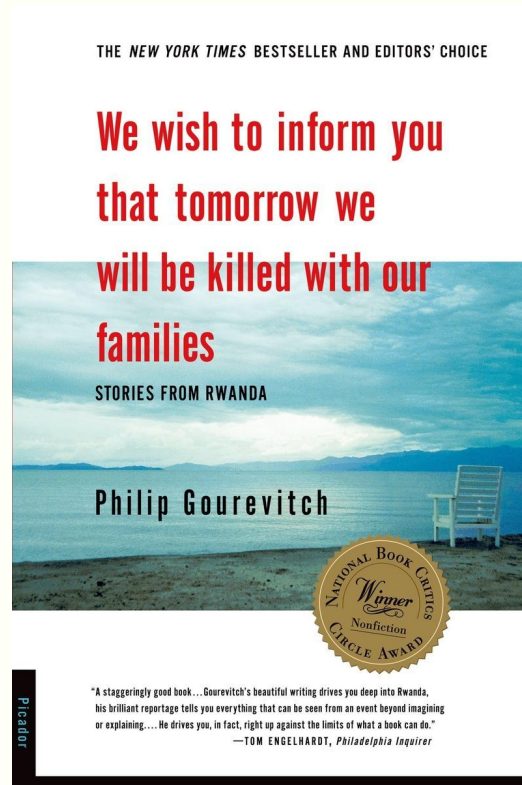
DATA MINIMIZATION

Collect **only what the stated purpose requires**, keep it only as long as needed, at the coarsest resolution that still serves the purpose.
(Core principle of the ICRC *Handbook on Data Protection in Humanitarian Action*.)



LiGHT

Laboratory for intelligent
Global Health & Humanitarian
Response Technologies.



End

DOMAIN

Questions?

Light

Laboratory for intelligent
Global Health & Humanitarian
Response Technologies.

Thank You

Except where otherwise indicated, original content in this presentation is © 2026 Hartley, Keitel, Klein, Sasu, Arni, Kesmen, Nemo and licensed under CC BY 4.0: <https://creativecommons.org/licenses/by/4.0/>.

Third-party materials are excluded from this license and remain subject to their indicated rights.

